

Olatomiwa Aluko

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Education

University of Central Florida (UCF)

Bachelor of Science in Computer Engineering, Minor in Technology Entrepreneurship

Orlando, FL

Expected May 2027

- Organizations: Colorstack, National Society of Black Engineers, Knight Hacks, Alpha Phi Alpha Fraternity, Inc.

Skills

Languages: Python, Java, C, HTML, CSS, JavaScript, TypeScript

Frameworks: ReactJS, Node.js, Next.js, NestJS, Tailwind CSS, tRPC, FastAPI

Developer Tools: Git/GitHub, Vercel, Postman Agent, Resend, Prisma, Supabase

Technologies: REST APIs, PostgreSQL, MySQL, Docker, Redis, Generative AI (Claude, ChatGPT, Gemini), SolidWorks, Microsoft 365, Adobe CC

Projects

NSBE UCF Event Tracker — Organization Management Platform

Next.js (React), TypeScript, NestJS, Prisma, PostgreSQL, OAuth, Railway, Vercel, Git/Github

- Architected full-stack event management system serving **100+ users**, implementing role-based access control with OAuth and RESTful APIs to streamline attendance tracking and event coordination for 50+ annual events.
- Reduced manual attendance processing time by **75%** with real-time check-in functionality, automated validation, and analytics dashboards leveraging Prisma ORM and PostgreSQL for optimized query performance.
- Deployed scalable microservices architecture on Railway and Vercel with **99.9% uptime**, implementing CI/CD pipelines and Docker containerization to handle **1,000+** concurrent database operations during peak usage.

CivicLens — Political Data Transparency Platform (Hackathon)

Next.js (React), TypeScript, FastAPI, PostgreSQL, Docker, Google Gemini AI, Git/Github, Mapbox GL, react-force-graph

- Led a team of 4 engineers** to develop a civic transparency platform democratizing political data access for **535+ congressional representatives**, implementing a **RAG-powered chatbot with 99.2% citation accuracy** using Google Gemini 2.5 Flash, PostgreSQL with pgvector, and FastAPI backend.
- Improved legislative data accessibility by **85%** by aggregating FEC, Congress.gov, and OpenSecrets APIs into interactive **3D network graphs** and choropleth maps, implementing fuzzy matching algorithms to resolve politician identities across disparate government databases.
- Architected responsible AI verification pipeline with semantic search and fact-checking mechanisms to prevent hallucinations; deployed with **three-tier caching strategy** reducing API calls by 85% and response times from 3s to 300ms.

Chalk — NBA Statline Prediction System

Python, XGBoost, FastAPI, PostgreSQL, React/TypeScript, Redis, Airflow, MLflow, Railway/Supabase, Git/Github

- Built a machine learning system using **XGBoost and LightGBM** trained on **147,000+ NBA game logs** to predict player statistics with accuracy comparable to Vegas sportsbooks, using 74 engineered features spanning rolling averages, opponent matchups, and player usage trends.
- Orchestrated a full data pipeline with **Apache Airflow**, running **2–3 DAGs daily via cron jobs** on Railway to handle game and odds ingestion in the morning, generate fresh player predictions in the evening, and run model drift checks overnight to ensure prediction accuracy stays consistent.
- Deployed a full-stack web application with **Redis caching and a 15-minute TTL refresh cycle**, featuring live stat distribution charts and a betting edge calculator tracking **523 active players** across a **12,900+ game** database.

Professional Experience

BNY

Summer 2026

Software Engineering Intern (Incoming)

- Selected for a competitive Software Engineering internship at a global financial technology and investment services company.

Handshake

Oct 2025 – Present

AI Research, Evaluation & Data Annotation (Contract)

- Conduct quality assurance, model evaluation, and data annotation work supporting the development of large-scale AI systems across diverse modalities.
- Collaborate cross-functionally to ensure consistency, accuracy, and adherence to established quality standards, contributing to improved system reliability and performance through rigorous review processes.